

AYUSH KASHYAP

AI Engineer | Machine Learning | LLM Systems | Agentic AI

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PROFESSIONAL SUMMARY

CS student at Bennett University (Expected May 2027) with deep expertise in Modern AI Engineering. Specialized in production-ready LLM applications, advanced RAG systems, and autonomous AI agents. 500+ hours of intensive coursework in cutting-edge AI technologies.

CORE COMPETENCIES

Generative AI & LLMs: Fine-tuning (LoRA/QLoRA) • RAG Systems • AI Agents • Prompt Engineering • Multimodal AI
Frameworks: LangChain • LangGraph • HuggingFace • FastAPI • Gradio • Docker • AWS
Advanced RAG: Semantic chunking • Hybrid search • Corrective/Adaptive RAG • Agentic RAG • GraphRAG • Guardrails
Classical ML: Scikit-learn • XGBoost • Random Forest • Data preprocessing • Feature engineering
Deep Learning: Transformers • PyTorch • Attention mechanisms • CNNs • RNNs • Backpropagation

EDUCATION

Bennett University | B.Tech Computer Science | Expected Graduation: May 2027

ADVANCED COURSES COMPLETED (2024-2025)

 Ed Donner LLM Engineering Bootcamp (8 weeks) •  Krish Naik Advanced RAG Course (4 weeks) •  Deep Learning & Transformers Specialization •  Claude Code & Deep Agents

KEY PROJECTS

- AI Knowledge Worker (RAG System)** — Production RAG with semantic search, vector embeddings, conversational memory | LangChain, FAISS, ChromaDB
- Trading Floor Multi-Agent System** — 4 autonomous agents × 44 tools × 6 MCP servers | LangGraph, FastAPI, React
- Fine-Tuned LLaMA vs GPT** — LoRA/QLoRA fine-tuning competing with frontier models at 100x lower cost | Weights & Biases
- Multimodal Airline Support Agent** — Autonomous agent with function calling, structured outputs, tool orchestration | Gradio UI
- Python to C++ Converter** — AI agent achieving 60,000x performance improvements | GPT-4/5 code generation
- Meeting Minutes Extractor** — Audio processing pipeline with transcript generation | Whisper, OpenAI, HuggingFace
- Price Prediction Capstone** — Multi-model comparison: traditional ML vs neural nets vs LLMs | PyTorch, XGBoost
- AI Brochure Generator** — Website scraping + LLM chaining + document generation | LangChain, Web APIs

ADVANCED EXPERTISE

RAG (Advanced): Semantic chunking • Hybrid search • MMR • HyDE • Query expansion • Corrective/Adaptive/Agentic RAG • Multimodal RAG • Vectorless RAG • GraphRAG with Neo4j
Agents & Agentic AI: LangGraph workflows • ReAct architecture • Multi-agent collaboration • Tool calling • MCP protocol • A2A communication • Human-in-the-loop
LLM Fine-tuning: LoRA/QLoRA • Dataset curation • Hyperparameter optimization • Training monitoring • Weights & Biases tracking

ADDITIONAL INFORMATION

Languages: Python, C++, SQL | **Status:** Open to internships & full-time roles | **Focus:** AI Engineering, RAG Systems, Agentic AI, Production ML

Last Updated: January 2025